Advanced Microeconomics

Dr. Juliane Hennecke
Nils Sulzmann

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Problem Set 3: Theory recap, Applications of Expenditure Functions

1. Theory

Explain the following terms:

a) Carte Blanche Principle.

b) Willingness to pay

c) Internal and external validity.

2. Application

1) There is an individual with utility function $U(x_1, x_2) = \sqrt{x_1} + x_2$. Wealth is given by $w > 0$ and price are $p_1, p_2 > 0$.

a) Determine the optimal consumption, the indirect utility function and the expenditure function.

b) Assume that the prices are given by $p_1 = p_2 = 3$. The income is $w = 120$ monetary units. Calculate the demanded quantities x_1 and x_2 and the corresponding utility level.

2) The expenditure function of an individual is given by $e(p, u) = u\sqrt{p_1 p_2}$. Accordingly, there are two consumption goods whose consumption quantities are denoted by x_1 and x_2 and whose prices are denoted by p_1 and p_2, respectively.

a) Calculate the indirect utility function.

b) Determine how much of x_1 and x_2 the individual consumes in the optimum as a function of prices and income.

c) Assume that the parameters are determined by the following values $(p_1^0, p_2^0, w^0) = (3, 3, 120)$ Determine the quantities demanded and the associated utility level.

d) Present the results in a suitable graphic.